

## SPARTA FORT MCCOY, WI, USA

WMO: 726437

Lat: 43.967N

Lon: 90.733W

Elev: 829

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 94940

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -15.0      | -8.1 | -21.3                         | 1.7 | -12.0 | -15.3 | 2.5 | -6.1 | 18.9                  | 22.1 | 17.0 | 20.6 | 1.0                   | 260  | 0.528 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 24.9                   | 90.1            | 73.1 | 87.7 | 71.4 | 83.8 | 69.5 | 75.4                | 85.6 | 73.5 | 83.3 | 71.8 | 80.5 | 8.7                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 72.4                            | 123.6 | 81.1 | 70.4 | 115.5 | 78.9 | 68.7 | 108.8 | 77.2 | 39.5          | 85.8 | 37.6 | 83.0 | 36.1 | 82.1 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |  |  |
| 18.3              | 16.3 | 14.0 | -25.6                      | 93.8 | 5.5                | 4.0 | -29.6                                              | 96.6 | -32.8      | 99.0 | -35.9      | 101.2 | -39.9      | 104.1 |  |  |
|                   |      |      | -25.9                      | 78.9 | 5.4                | 2.1 | -29.8                                              | 80.4 | -33.0      | 81.6 | -36.0      | 82.8  | -40.0      | 84.3  |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 45.0        | 17.3  | 19.4  | 32.5  | 45.1  | 56.1  | 65.7  | 69.9  | 67.3  | 60.0  | 47.2  | 35.3  | 22.8  | (6)  |
|                                                                        | DBStd       | 20.87       | 13.31 | 12.43 | 12.41 | 10.01 | 8.79  | 6.68  | 6.09  | 6.18  | 8.40  | 9.26  | 10.21 | 11.54 | (7)  |
|                                                                        | HDD50       | 4144        | 1014  | 857   | 558   | 209   | 35    | 0     | 0     | 0     | 12    | 167   | 450   | 842   | (8)  |
|                                                                        | HDD65       | 7747        | 1479  | 1277  | 1007  | 601   | 301   | 68    | 21    | 45    | 194   | 557   | 890   | 1307  | (9)  |
|                                                                        | CDD50       | 2339        | 0     | 0     | 17    | 62    | 226   | 473   | 617   | 537   | 314   | 82    | 10    | 1     | (10) |
|                                                                        | CDD65       | 462         | 0     | 0     | 1     | 3     | 26    | 91    | 173   | 117   | 45    | 6     | 0     | 0     | (11) |
|                                                                        | CDH74       | 5537        | 0     | 0     | 10    | 70    | 380   | 1101  | 2073  | 1285  | 548   | 70    | 0     | 0     | (12) |
|                                                                        | CDH80       | 1673        | 0     | 0     | 0     | 12    | 99    | 316   | 743   | 364   | 131   | 8     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 5.1         | 5.1   | 5.3   | 5.8   | 6.9   | 5.9   | 4.8   | 4.2   | 3.8   | 4.3   | 5.1   | 5.2   | 5.0   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 35.0        | 1.1   | 0.9   | 1.9   | 3.5   | 4.4   | 5.2   | 4.4   | 4.8   | 3.4   | 2.4   | 1.9   | 1.2   | (15) |
|                                                                        | PrecMax     | 44.3        | 2.6   | 1.8   | 3.3   | 6.4   | 10.8  | 9.4   | 9.3   | 15.0  | 9.2   | 5.8   | 5.3   | 2.6   | (16) |
|                                                                        | PrecMin     | 24.8        | 0.4   | 0.1   | 0.3   | 1.5   | 1.7   | 1.8   | 0.9   | 1.8   | 0.6   | 0.7   | 0.3   | 0.4   | (17) |
|                                                                        | PrecStd     | 5.1         | 0.6   | 0.5   | 0.9   | 1.5   | 2.0   | 2.6   | 2.0   | 2.9   | 2.1   | 1.3   | 1.2   | 0.7   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 48.5  | 55.1  | 73.1  | 81.0  | 88.4  | 91.0  | 95.4  | 91.5  | 88.5  | 81.0  | 69.7  | 54.6  | (19) |
|                                                                        |             | MCWB        | 41.0  | 49.4  | 60.2  | 61.0  | 69.2  | 73.6  | 73.4  | 74.5  | 72.2  | 66.8  | 56.3  | 50.7  | (20) |
|                                                                        | 2%          | DB          | 42.9  | 46.1  | 63.8  | 74.5  | 82.0  | 87.9  | 90.6  | 88.0  | 83.7  | 74.8  | 62.1  | 46.3  | (21) |
|                                                                        |             | MCWB        | 37.3  | 40.6  | 53.2  | 57.2  | 64.3  | 71.0  | 73.7  | 71.9  | 69.9  | 61.4  | 52.8  | 42.6  | (22) |
|                                                                        | 5%          | DB          | 37.1  | 40.8  | 57.4  | 69.9  | 78.4  | 83.6  | 87.9  | 84.0  | 80.5  | 69.9  | 56.6  | 41.9  | (23) |
|                                                                        |             | MCWB        | 34.5  | 36.4  | 48.6  | 55.1  | 62.5  | 68.5  | 72.3  | 70.2  | 66.7  | 60.0  | 49.0  | 38.4  | (24) |
|                                                                        | 10%         | DB          | 35.5  | 36.5  | 52.0  | 64.3  | 73.4  | 80.6  | 84.0  | 81.0  | 76.9  | 64.0  | 53.1  | 37.4  | (25) |
|                                                                        |             | MCWB        | 33.3  | 33.0  | 44.5  | 50.9  | 60.2  | 66.1  | 70.3  | 68.0  | 65.3  | 56.0  | 46.6  | 34.8  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 42.3  | 46.9  | 61.6  | 63.7  | 72.0  | 76.5  | 78.9  | 76.7  | 74.3  | 71.3  | 58.0  | 52.5  | (27) |
|                                                                        |             | MCDB        | 48.1  | 55.2  | 73.1  | 76.6  | 84.8  | 87.9  | 89.0  | 86.4  | 85.3  | 78.2  | 64.7  | 55.7  | (28) |
|                                                                        | 2%          | WB          | 37.9  | 41.0  | 56.2  | 59.4  | 68.1  | 73.1  | 76.4  | 74.5  | 71.6  | 65.2  | 54.3  | 43.1  | (29) |
|                                                                        |             | MCDB        | 41.7  | 45.6  | 61.9  | 72.0  | 77.8  | 83.7  | 86.8  | 83.3  | 80.1  | 71.9  | 60.8  | 45.4  | (30) |
|                                                                        | 5%          | WB          | 35.1  | 36.6  | 49.4  | 56.1  | 65.5  | 70.9  | 74.0  | 72.6  | 69.2  | 60.8  | 50.3  | 38.5  | (31) |
|                                                                        |             | MCDB        | 37.3  | 40.4  | 57.0  | 66.5  | 74.6  | 80.3  | 84.3  | 80.4  | 76.7  | 68.4  | 56.0  | 41.4  | (32) |
|                                                                        | 10%         | WB          | 32.8  | 33.3  | 44.2  | 52.5  | 62.3  | 68.5  | 72.1  | 70.7  | 67.0  | 56.7  | 46.3  | 35.4  | (33) |
|                                                                        |             | MCDB        | 34.5  | 35.8  | 50.7  | 62.7  | 70.9  | 77.1  | 81.2  | 78.2  | 74.4  | 63.5  | 51.3  | 37.6  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 18.7  | 20.8  | 21.1  | 23.6  | 24.9  | 23.9  | 24.9  | 23.9  | 24.9  | 22.1  | 18.5  | 16.6  | (35) |
|                                                                        |             | MCDBR       | 18.7  | 23.2  | 26.8  | 32.2  | 29.6  | 27.7  | 27.5  | 26.6  | 27.5  | 28.7  | 26.4  | 20.4  | (36) |
|                                                                        | 5% WB       | MCWBR       | 15.0  | 17.9  | 16.1  | 18.1  | 14.8  | 13.9  | 13.0  | 13.0  | 14.7  | 17.3  | 18.4  | 16.7  | (37) |
|                                                                        |             | MCWBR       | 17.9  | 21.8  | 22.6  | 26.6  | 24.7  | 23.1  | 24.0  | 21.8  | 23.3  | 24.5  | 23.9  | 17.8  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.262 | 0.290 | 0.325 | 0.365 | 0.393 | 0.404 | 0.403 | 0.405 | 0.365 | 0.329 | 0.295 | 0.265 | (40) |
|                                                                        |             | taud        | 2.400 | 2.320 | 2.374 | 2.333 | 2.302 | 2.301 | 2.314 | 2.309 | 2.411 | 2.502 | 2.516 | 2.458 | (41) |
|                                                                        | Ebn at Noon |             | 279   | 286   | 290   | 285   | 279   | 275   | 274   | 269   | 272   | 270   | 265   | 267   | (42) |
|                                                                        |             | Edh at Noon | 23    | 31    | 33    | 38    | 40    | 41    | 40    | 38    | 32    | 26    | 21    | 20    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 561   | 910   | 1241  | 1432  | 1704  | 1884  | 1948  | 1656  | 1329  | 850   | 553   | 412   | (44) |
|                                                                        | RadStd      |             | 63    | 83    | 148   | 135   | 131   | 145   | 95    | 111   | 100   | 108   | 48    | 44    | (45) |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (49 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page